MATH 1075 - Section 9.1 Worksheet

10/20/2026

1 Intersections and Unions of Sets

1. Given two sets $P = \{a, b, c, d, e, f, g, h, i\}$ and $Q = \{a, e, i, o, u\}$, find

   a) $P \cup Q$

   b) $P \cap Q$

2. Given sets $A = \{x \mid x < -4\}$, $B = \{x \mid x > 2\}$ and $C = \{x \mid 0 \leq x \leq 5\}$, find

   a) $A \cup B$

   b) $A \cap B$

   c) $A \cup C$

   d) $A \cap C$

   e) $B \cup C$

   f) $B \cap C$

3. Find unions and intersections of sets as indicated.

   a) $(-\infty, 4) \cap [-1, 5)$

   b) $(-\infty, 4) \cup [-1, 5)$

2 Solving Compound Inequalities: And

Solve the compound inequality and graph the solution. Write your answer in interval notation.

1. $x + 6 < -2$ and $5x < 30$

2. $5x + 2x \geq -21$ and $-9x + 3x \geq -24$

3. $\frac{5}{2}(x + 2) < -6$ and $\frac{3}{4}(x - 2) < 1$

4. $-8 \geq -3x - 2$ and $3(x - 7) + 16 > 4x$

3 Solving Inequalities of the Form $a < x < b$

Solve the inequality and graph your solution. Write your answer in interval notation.

1. $0 \leq 2x - 5 < 9$

2. $-6 < 3x - 9 \leq 0$

3. $\frac{1}{3} > \frac{x-4}{-3} > -2$

4. $12 > 6x + 3 \geq 0$

4 Solving Compound Inequalities: Or

Solve the compound inequality and graph the solution. Write your answer in interval notation.

1. $x < 0$ or $3x + 1 \geq 7$

2. $22 > 4x - 10$ or $7 > 2x - 5$

3. $\frac{3}{8}x + 1 > 0$ or $-2x \geq -4$

4. $1.25x + 3 \leq 0.5x - 6$ or $2.5x - 1 \geq 9 - 1.5x$